

MH1100 Calculus I

Final Review

Final Review

Limit:

- Definition ($\epsilon - \delta$ problem)

(1) Prove $\lim_{x \rightarrow 5} x(x-5) = 0$; (2) Prove $\lim_{x \rightarrow \infty} \frac{1}{x} = 0$. $\epsilon - N$

- Limit laws $\lim_{x \rightarrow 0^-} = \lim_{x \rightarrow 0^+}$

(1) $\lim_{x \rightarrow 0} \frac{|x|}{x}$; (2) $\lim_{x \rightarrow 0} \frac{(x^3+27)^{1/3}-3}{x}$. $\Rightarrow a^3 - b^3 = (a-b)(a^2 + ab + b^2)$

- Limit at infinity (Note that ∞ is not a number).

(1) $\lim_{x \rightarrow \infty} \sin(1/x)$.

$\lim_{x \rightarrow \infty} x^2 = \infty$

- Special limit

(1) $\lim_{x \rightarrow 0} \frac{\sin(x^5+2x^3)}{x^3}$; (2) $\lim_{x \rightarrow 0} (1+x)^{\frac{1}{x}} = e$

$\lim_{n \rightarrow \infty} (1 + \frac{1}{n})^n = e$

$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$

- Squeeze Theorem

(1) $\lim_{x \rightarrow 0} x \sin(1/x)$.

$g(x) \leq f(x) \leq h(x) \quad \lim_{x \rightarrow a} g(x) = \lim_{x \rightarrow a} h(x)$

Final Review

Continuity:

$$\lim_{x \rightarrow a} f(x) = f(a)$$

- note
- ① limit exist
 - ② $f(a)$ exist
 - ③ $\lim = \text{function}$

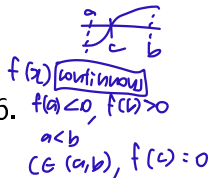
- Definition

(1) Prove that $f(x) = |x|$ is continuous in the region $[-10, 10]$.

- The Intermediate Value Theorem

(1) Show that there is a root of the equation

$$x^{201} - 5x^{200} + 6x^2 - 21x - 46 = 0 \text{ between 5 and 6.}$$



Asymptote

- Vertical asymptote $x=a$ $\lim_{x \rightarrow a^+ / a^-} f(x) = +\infty / -\infty$

(1) What is the vertical asymptote for function $f(x) = \frac{2 \tan(x)}{(1+x)(x-5)}$.

- Horizontal asymptote $y=a$ $\lim_{x \rightarrow \infty / -\infty} f(x) = a$

(1) What is the horizontal asymptote for function $f(x) = \frac{\sqrt{2x^2+10}}{\sqrt{x^3+x^2+10}}$.

- Slant asymptote $y=mx+b$ $\lim_{x \rightarrow \infty / -\infty} f(x) - (mx+b) = 0$
 $m \neq 0$

(1) What is the slant asymptote for function $f(x) = \frac{3x^5+x^4+10}{6x^4+x^3+2}$.

Final Review

Derivative:

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

- Definition

(1) Prove that $f'(x) = 3x^2$ for $f(x) = x^3$;

- Higher order derivative

(1) $f(x) = \frac{x \sin(x)}{x + \cos(x)}$, find $f''(x)$ $f'(a), f''(a), f'''(a), \dots, f^{(n)}(a)$

- Differentiation formulas and chain rules ★

(1) $f(x) = \frac{x^2 + \tan(x)}{\sin(x)}$;

(2) $f(x) = \frac{\sin(x^2 \cos(x^2 + \tan x)) + x^3 \sec x}{y^2 f(x)}$

- Implicit differentiation

(1) Find the tangent to $x^3 + y^3 = 6xy$ at the point $(3, 3)$, and find y'' at this point $\frac{d}{dx} \quad \Downarrow \quad y(x) \quad \frac{d}{dx}$

- linear approximation

(1) Estimate the given number $(1002)^{1/3}$ $f(x) \approx L(x) = f(a) + f'(a)(x-a)$

Final Review

Application of derivative:

$$\begin{cases} S(t) = 4\pi r^2(t) & \frac{dS}{dt} = \frac{dr}{dt} \\ v(t) = \frac{4\pi}{3} r^3(t) & \frac{dv}{dt} = \frac{dr}{dt} \end{cases}$$

- Rates of Change

(1) Air is being pumped into a spherical balloon so that its surface area increases at a rate of 100 cm^2 . How fast is the radius of the balloon increasing when the diameter is 10 cm?

- Maximum and minimum values

global max and min only @ $\begin{cases} \textcircled{1} \text{ end point values} \\ \textcircled{2} \text{ at critical number} \end{cases}$

(1) Find the absolute maximum and absolute minimum of the function $f(x) = x^3 - x^2 - x$ at $[0, 2]$;

- Increasing/Decreasing and concave upward and downward

(1) Prove that $\sin x < x < \tan x$ for $0 < x < 1$; $\rightarrow \text{study } f''(x)$

(2) Find the intervals of concavity and the inflection points for function $f(x) = 2x^3 + 3x^2 + x + 8$

- The Mean Value Theorem, Rolle's Theorem, Cauchy's Mean Value Theorem

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

(1) Prove that inequality $\sin x - \sin y \leq |x - y|$ for all x and y .

Final Review

Application of derivative:

- Sketch a function

(1) Sketch the function $f(x) = \frac{x^2}{x^2 - 5x + 6}$

Check:
① odd/even function
② intercepts
③ increasing/decreasing
④ concavity
⑤ asymptotes

- Optimization Problems

(1) Find the area of the largest rectangle that can be inscribed in a circle of radius r .

- Newton's Method $\Rightarrow x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$ $f'(x_n) \neq 0$

(1) Starting with $x_1 = 2$, find the third approximation x_3 to the root of the equation $x^3 - 2x - 5 = 0$.

- Antiderivatives $\rightarrow$ Integration

(1) Find f if $f''(x) = 12x^2 + 6x - 4$, $f(0) = 4$, and $f(1) = 1$

$$f(x) + C$$

Final Review

Inverse Function

$$b^x \quad \nwarrow \quad \nearrow \log_b b^x$$

$(e^x)' = e^x \rightarrow \log_b b^x$

- Inverse Functions

(1) Find $(f^{-1})'(a)$ for $f(x) = 2x^3 + 3x^2 + 7x + 4$, here $a = 4$.

- Derivatives for Exponential Functions and Logarithmic Functions

(1) Differentiate the function (a) $y = e^{\tan(x)}$; (b) $y = e^{t \sin(t)}$.

(2) Differentiate the function (a) $y = \frac{\ln x}{1 + \ln x}$; (b) $y = (\ln x)^{\sin x}$.

- L'Hospital's Rule $\frac{0}{0}, \frac{\infty}{\infty}, 0 \cdot \infty, \frac{0}{\infty}$

(1) $\lim_{x \rightarrow \infty} \frac{\ln x}{e^x - 1 - x}$

(2) $\lim_{x \rightarrow 0} \frac{e^x - 1 - x}{x^2}$

(3) $\lim_{x \rightarrow 0} (1 + \sin 3x)^{1/x}$

$$\downarrow$$
$$\ln y = \ln [(\ln x)^{\sin x}]$$
$$= \sin x \ln(\ln x)$$

$$(1) \lim_{x \rightarrow \infty} \frac{(\ln x)^n}{b^x} = 0$$

$(\ln x)^n \ll x^p \ll b^x$
 $x \rightarrow \infty$

Final Review with Hints

Hints for Limit:

- Definition ($\epsilon - \delta$ problem)

(1) Prove $\lim_{x \rightarrow 5} x(x - 5) = 0$; (2) Prove $\lim_{x \rightarrow \infty} \frac{1}{x} = 0$.

Hint: (1) Let x be $4 < x < 6$, so we have $|x| < 6$. We can let $\delta < \epsilon/6$, $|x(x - 5) - 0| < 6|x - 5| < \epsilon$.

(2) Use $N = 1/\epsilon$, let $\epsilon > 0$, we can have $N = 1/\epsilon$.

- Limit laws

(1) $\lim_{x \rightarrow 0} \frac{|x|}{x}$; (2) $\lim_{x \rightarrow 0} \frac{(x^3 + 27)^{1/3} - 3}{x}$.

Hint: (1) D.N.E

(2) Let $f(x) = (x^3 + 27)^{1/3}$, so $f'(0) = \lim_{x \rightarrow 0} \frac{(x^3 + 27)^{1/3} - 3}{x}$. We have $f'(0) = 0$.

Final Review with Hints

Solution for Limit:

- Limit at infinity (Note that ∞ is not a number).

(1) $\lim_{x \rightarrow \infty} \sin(1/x)$.

Hint: Substitute $t = 1/x$. Then $t \rightarrow 0^+$ as $x \rightarrow \infty$.

- Special limit

(1) $\lim_{x \rightarrow 0} \frac{\sin(x^5+2x^3)}{x^3}$; (2) $\lim_{x \rightarrow 0} (1+x)^{\frac{1}{x}} = e$

Hint: (1) $\lim_{x \rightarrow 0} \frac{\sin(x^5+2x^3)}{x^3} = \lim_{x \rightarrow 0} \frac{\sin(x^5+2x^3)}{x^5+2x^3} \frac{x^5+2x^3}{x^3} = 2$

- Squeeze Theorem

(1) $\lim_{x \rightarrow 0} x \sin(1/x)$.

Hint: $-|x| \leq x \sin(1/x) \leq |x|$

Continuity:

- Definition

(1) Prove that $f(x) = |x|$ is continuous in the region $[-10, 10]$.

Hint: $f(x) = x, x > 0; f(x) = 0, x = 0; f(x) = -x, x < 0$.

- The Intermediate Value Theorem

(1) Show that there is a root of the equation

$x^{201} - 5x^{200} + 6x^2 - 21x - 46 = 0$ between 5 and 6.

Hint: $f(x) = x^{201} - 5x^{200} + 6x^2 - 21x - 46$ is continuous. We have $f(5) * f(6) < 0$.

Final Review with Hints

Asymptote

- Vertical asymptote

(1) What is the vertical asymptote for function $f(x) = \frac{2 \tan(x)}{(1+x)(x-5)}$.

Hint: $x = -1$, $x = 5$, and $x = \pi/2 + N * \pi$ (N is any integer)

- Horizontal asymptote

(1) What is the horizontal asymptote for function $f(x) = \frac{\sqrt{2x^2+10}}{\sqrt{x^3+x^2+10}}$.

Hint: $y = 0$

- Slant asymptote

(1) What is the slant asymptote for function $f(x) = \frac{3x^5+x^4+10}{6x^4+x^3+2}$.

Hint: $y = \frac{1}{2}x + \frac{1}{12}$

Final Review with Hints

Derivative:

- Definition

(1) Prove that $f'(x) = 3x^2$ for $f(x) = x^3$;

Hint: $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{(x+h)^3 - x^3}{h}$

- Higher order derivative

(1) $f(x) = \frac{x \sin(x)}{x + \cos(x)}$, find $f''(x)$

Hint: Use quotient rule twice.

- Differentiation formulas and chain rules

(1) $f(x) = \frac{x^2 + \tan(x)}{\sin(x)}$;

(2) $f(x) = \sin(x^2 \cos(x^2 + \tan x)) + x^3 \sec x$

Hint: Use quotient rule and chain rule.

Final Review with Hints

Derivative:

- Implicit differentiation

(1) Find the tangent to $x^3 + y^3 = 6xy$ at the point $(3, 3)$, and find y'' at this point

Hint: (1) $3x^2 + 3y^2y' = 6y + 6xy'$, so $y' = \frac{x^2 - 2y}{2x - y^2}$. At $(3,3)$, $y' = -1$

From $x^2 + y^2y' = 2y + 2xy'$, we have $2x + 2yy' - 4y' = (2x - y^2)y''$, we have $y'' = -\frac{4}{3}$.

- linear approximation

(2) Estimate the given number $(1002)^{1/3}$

Hint: (2) Let $f(x) = x^{1/3}$, $a = 1000$, and $f(x) \approx f(a) + f'(a)(x - a)$.

Final Review with Hints

Application of derivative:

- Rates of Change

(1) Air is being pumped into a spherical balloon so that its surface area increases at a rate of 100cm^2 . How fast is the radius of the balloon increasing when the diameter is 10 cm?

Hint: $S = 4\pi r^2$, $dS = 8\pi r dr$, and diameter = $2 \times$ radius.

- Maximum and minimum values

(1) Find the absolute maximum and absolute minimum of the function $f(x) = x^3 - x^2 - x$ at $[0, 2]$;

Hint: $f'(x) = 3x^2 - 2x - 1 = (x - 1)(3x + 1)$, check $f(0)$, $f(1)$, $f(2)$.

Note that $x = -1/3$ is not in the region $[0, 2]$.

Final Review with Hints

Application of derivative:

- Increasing/Decreasing and concave upward and downward

(1) Prove that $\sin x < x < \tan x$ for $0 < x < \frac{\pi}{2}$;

Hint: define $f(x) = x - \sin x$ and $g(x) = \tan x - x$, check $f'(x)$ and $g'(x)$.

(2) Find the intervals of concavity and the inflection points for function $f(x) = 2x^3 + 3x^2 + x + 8$

Hint: Find $f''(x)$

- The Mean Value Theorem, Rolle's Theorem, Cauchy's Mean Value Theorem

(1) Prove that inequality $\sin x - \sin y \leq |x - y|$ for all x and y .

Hint: Let $f(x) = \sin x$, so $f(x)$ is continuous and differential. We have $\sin x - \sin y = \cos(c)(x - y)$, here c is a number between x and y .

Final Review with Hints

Application of derivative:

- Sketch a function

(1) Sketch the function $f(x) = \frac{x^2}{x^2 - 5x + 6}$

Hint: Check domain, intercepts, symmetry, asymptotes, I/D test, Local max/min, concavity and points of inflection.

- Optimization Problems

(1) Find the area of the largest rectangle that can be inscribed in a circle of radius r .

Hint: Optimization problem (check lecture note)

- Newton's Method

(1) Starting with $x_1 = 2$, find the third approximation x_3 to the root of the equation $x^3 - 2x - 5 = 0$.

Hint: $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$

- Antiderivatives

(1) Find f if $f''(x) = 12x^2 + 6x - 4$, $f(0) = 4$, and $f(1) = 1$

Hint: calculate f' first, then from f' to f .

Inverse Function

- Inverse Functions

(1) Find $(f^{-1})'(a)$ for $f(x) = 2x^3 + 3x^2 + 7x + 4$, here $a = 4$.

Hint: $(f^{-1})'(a) = \frac{1}{f'(f^{-1}(a))}$

- Derivatives for Exponential Functions and Logarithmic Functions

(1) Differentiate the function (a) $y = e^{\tan(x)}$; (b) $y = e^{t \sin(t)}$.

Hint: (a) $\ln y = \tan x$, so $\frac{y'}{y} = \sec^2 x$;

(b) $\ln y = t \sin t$.

(2) Differentiate the function (a) $y = \frac{\ln x}{1 + \ln x}$; (b) $y = (\ln x)^{\sin x}$.

Hint: (a) $\ln y = \ln(\ln x) - \ln(1 + \ln x)$;

(b) $\ln y = \sin x \ln x$

Inverse Function

- L'Hospital's Rule

(1) $\lim_{x \rightarrow \infty} \frac{\ln x}{x}$

Hint: $\lim_{x \rightarrow \infty} \frac{\ln x}{x} = \lim_{x \rightarrow \infty} \frac{1/x}{1} = 0$

(2) $\lim_{x \rightarrow 0} \frac{e^x - 1 - x}{x^2}$

Hint: $\lim_{x \rightarrow 0} \frac{e^x - 1 - x}{x^2} = \lim_{x \rightarrow 0} \frac{e^x - 1}{2x} = \lim_{x \rightarrow 0} \frac{e^x}{2} = 1/2$ (3)

$\lim_{x \rightarrow 0} (1 + \sin 3x)^{1/x}$

Hint: Let $y(x) = (1 + \sin 3x)^{1/x}$, $\ln y(x) = 1/x \ln(1 + \sin 3x)$,

$\lim_{x \rightarrow 0} \ln y(x) = \lim_{x \rightarrow 0} \frac{\ln(1 + \sin 3x)}{x} = \lim_{x \rightarrow 0} \frac{3 \cos 3x}{1 + \sin 3x} = 3$, thus

$\lim_{x \rightarrow 0} (1 + \sin 3x)^{1/x} = e^3$.